Engine Model(s): PT6A

72-50-02

COMPRESSOR TURBINE - MAINTENANCE PRACTICES

1. General

A. Maintenance personnel should make reference to the INTRODUCTION section and Chapter 70-00-00 STANDARD PRACTICES of this manual to familiarize themselves with general procedures.

B. Install suitable protective caps/covers over all disconnected tubes/lines and component openings.

C. Lockwire shall comply with specification AMS 5687, heat and corrosion resistant steel wire MS9226-03, is 0.025 inch diameter, and will not be specified in instructions.

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name
PWC05-103	Marker, Pencil
PWC06-004	Compound, Anti-seize

3. Special Tools

The special tools listed below are used in the following procedures.

Tool No.	Name
PWC30331	Wrench
PWC30335	Spreader
PWC30336	Protector
PWC30403	Puller
PWC30458	Squeezer, Keywasher
PWC50057	Drill Kit
PWC50060	Injector

4. Fixtures, Equipment and Supplier Tools

Not Applicable

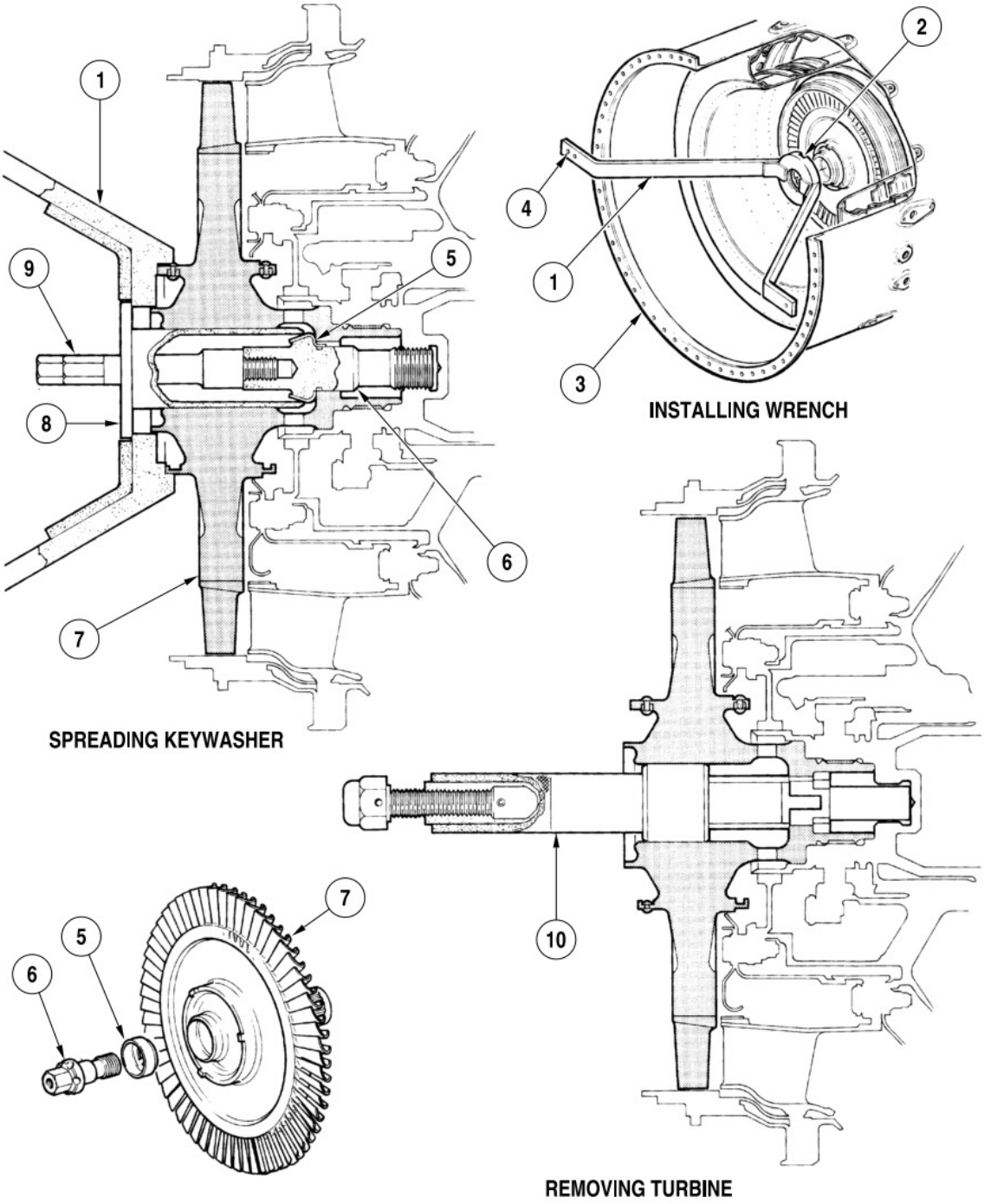
5. Removal/Installation

A. Removal of Compressor Turbine Disk Assembly (Ref. Figs. 201 and 202)

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Figure 201 Removal/Installation of Compressor Turbine Disk Assembly

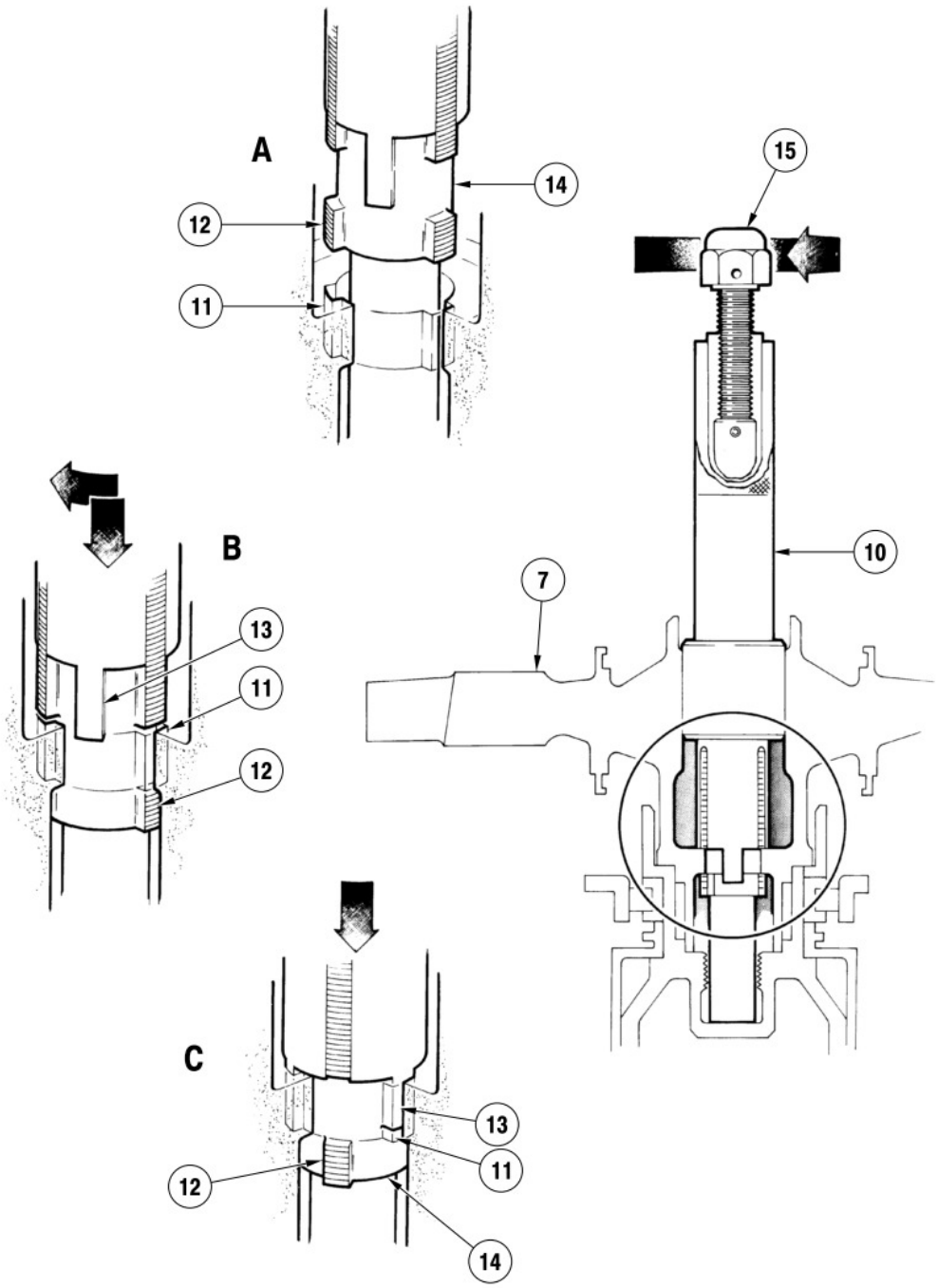
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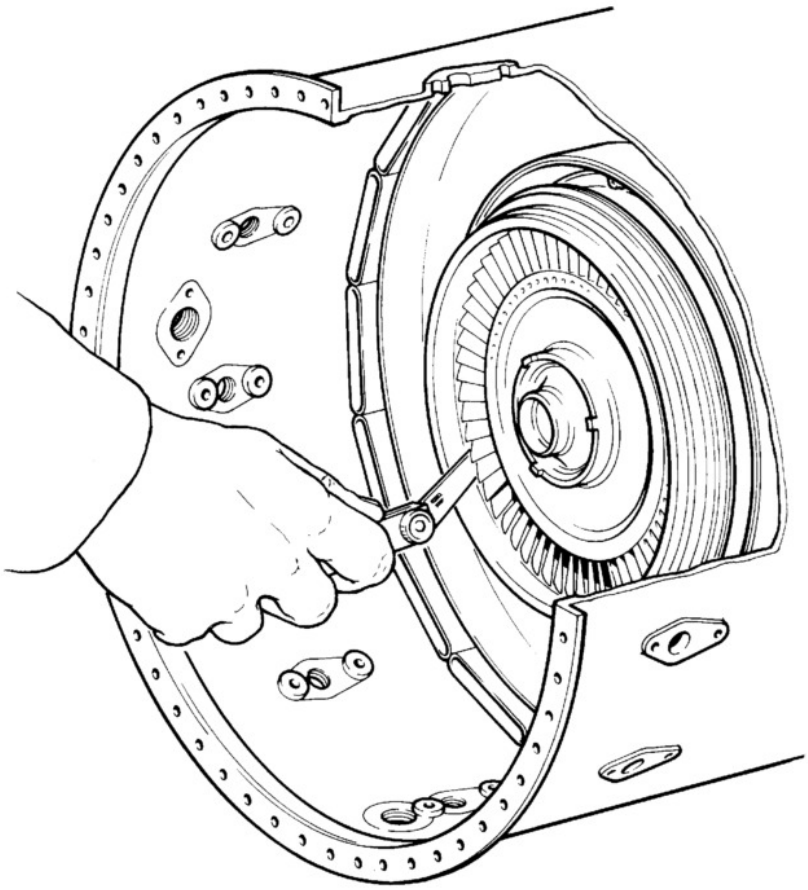
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COMPRESSOR TURBINE AND TIP CLEARANCE CHECK – TYPICAL

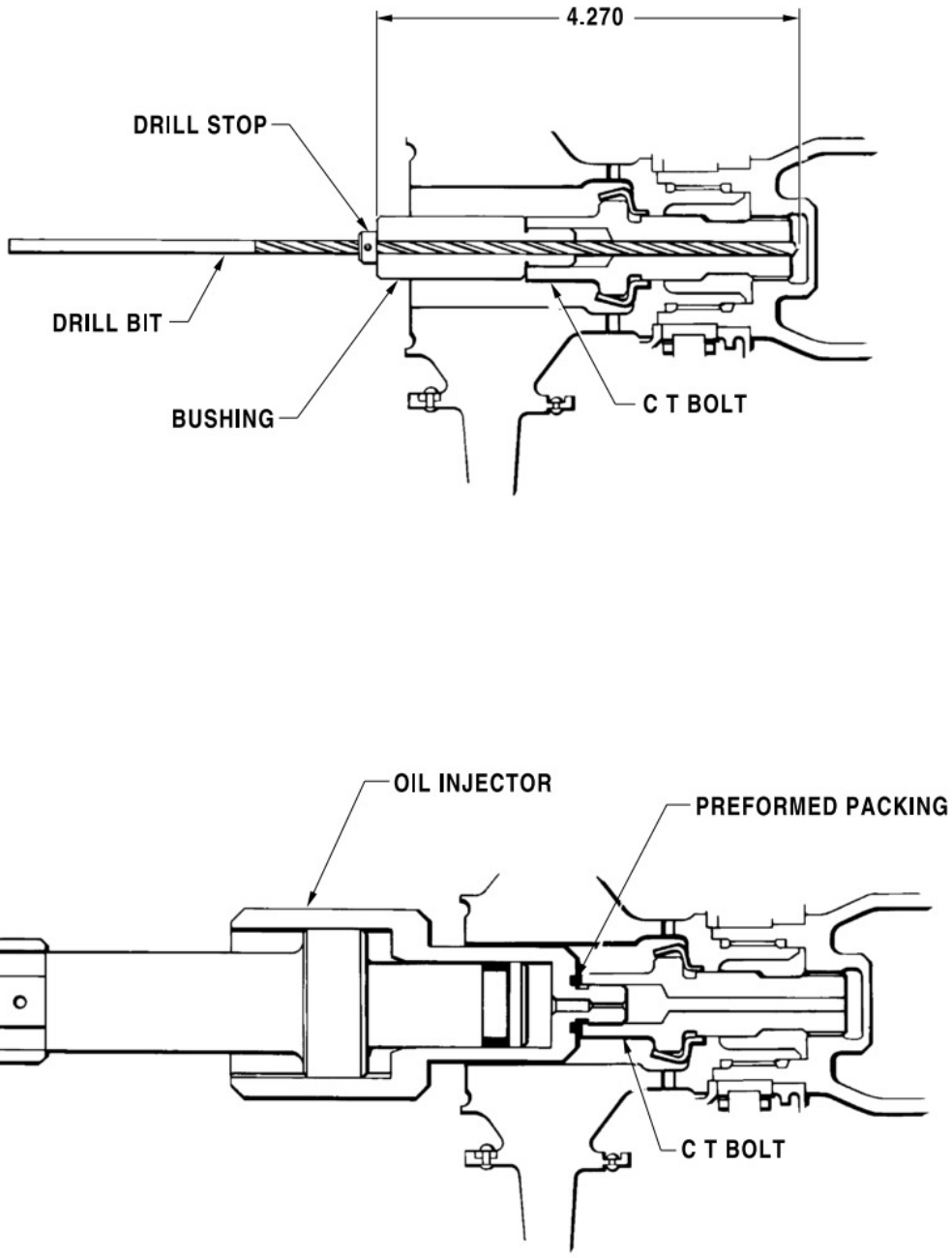
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1. Wrench (PWC30331)
2. Slot
3. Flange C
4. Locating Pin
5. Keywasher
6. Stubshaft Bolt
7. Compressor Turbine
8. Protector (PWC30336)
9. Spreader (PWC30335)
10. Puller (PWC30403)
11. Disk Slot
12. Lug
13. Lock
14. Puller Body
15. Centerscrew

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Figure 202 Removal of CT Bolt



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(1) Remove the power section and install in stand (Ref. 72-00-00, REMOVAL/INSTALLATION).

(2) Use wire or tapered feeler gages to measure the clearance of each blade at the 6 o'clock position and determine the longest blade. Put a mark on the longest blade using marker (PWC05-103).

NOTE: CT blades may be flat on the tip or may have a taper. For CT blades that are flat on the tip, measure the CT disk OD at the highest point of the blades (leading or trailing edge). For blades with a taper, measure the CT disk OD at the blade trailing edge.

(3) Using wire or feeler gages measure tip clearance over each blade. (Ref. Chap. 72-50-01, FIT. No. 318, Fits and Clearances).

NOTE: During clearance checks, turbine bladed disk should be loaded radially, by hand pressure, to obtain smallest possible gap.

(4) Check that the tip clearances are within acceptable limits (Ref. Chap. 72-50-01, FIT. No. 318, Fits and Clearances). If the tip clearances are not within limits, length of longest blade must be calculated and one or more shroud segments may require replacement or grinding (Ref. Chap. 72-50-01, Approved Repairs).

(5) Place wrench (1, (PWC30331), Fig. 201) on gas generator case with extension arm locating pins (4) inserted into bolt holes in Flange C (3). Turn compressor turbine (7) to engage disk lugs with wrench slots (2).

(6) Insert protector (8, PWC30336) into bore of wrench (1, PWC30331) and compressor turbine disk.

(7) Place spreader (9, PWC30335) with center bolt in protector (8, PWC30336). Screw the center bolt into thread in head of compressor turbine stubshaft bolt (6) until keywasher is unlocked. Remove the spreader.

(8) **Pre-SB1439**: Remove the stubshaft bolt as follows:

CAUTION: DO NOT TORQUE THE BOLT MORE THAN 150 LBF.FT.

(a) Apply breakaway torque of 50 to 100 lbf.ft. on the CT bolt with a standard socket and wrench (PWC30331) (1).

(b) If **Pre-SB1439** CT bolt is seized (Ref. Fig. 202):

1 Install the drill bit (part of kit PWC50057) in a suitable hand drill.

2 Install the drill stop on the drill bit 4.270 in. from tip.

3 Install the bushing into the CT bolt hole and tighten with your hand.

4 Use penetrating oil, and drill the bolt slowly until the drill stop contacts the bushing. Remove debris at regular intervals.

5 Remove the bushing from the CT bolt, and clean surrounding area.

6 Fill oil injector (PWC50060) as folows:

a Turn the knob clockwise until the plunger bottoms.

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b Place the injector tip in penetrating oil (50% kerosene, 50% engine oil is suitable).

c Turn the knob counterclockwise to fill the oil in the injector (PWC50060).

7 Install the injector (PWC50060) in the CT bolt. Make sure that the packing seals between the injector and bolt head is correct.

8 Turn the injector knob clockwise with your hand to empty the injector and to flood stubshaft cavity. Maintain pressure in cavity by tighten the knob at regular intervals for five minutes minimum.

9 Remove the injector (PWC50060), and loosen the CT bolt.

10 If CT bolt does not move, repeat Steps 6 thru 9 until bolt is removed.

NOTE: After the pilot hole has been drilled, discard **Pre-SB1439** CT bolt.

(9) **Post-SB1439**: Remove the CT bolt as follows:

(a) Apply sufficient amount of penetrating oil (50% kerosene, 50% engine oil is suitable) into CT bolt center hole to flood stubshaft cavity.

(b) Rotate the turbine several times to let oil to reach slots in bolt thread and let soak for 20 minutes minimum.

CAUTION: DO NOT TORQUE THE BOLT MORE THAN 150 LBF.FT.

(c) Apply breakaway torque of 50 to 100 lbf.ft. on the CT bolt. If the bolt does not move, apply sufficient torque enough to the move bolt, then loosen. Tighten/loosen the bolt until it is free and can be removed.

(10) If you are not able to remove the bolt after you apply penetrating oil, that indicates permanent seizure. Send the engine to an approved P&WC approved overhaul facility for repair.

(11) Remove protector (8, (PWC30336), Fig. 201), wrench (1, PWC30331), bolt (6) and keywasher (5).

(12) Insert puller (10, PWC30403) into compressor turbine bore to release turbine:

(a) View A: Engage lugs (12) with slots (11) in turbine disk.

(b) View B: Ease puller body (14) into turbine bore until lugs (12) are below slots (11).

(c) View C: Turn puller body (14) to position lugs (12) behind internal shoulder of disk.

(d) Push puller body (14) in to engine locks (13) in disk slots (11).

CAUTION: WHEN THE DISK IS RELEASED, DISCONTINUE TURNING THE CENTER SCREW TO AVOID COMPRESSOR TURBINE BLADES CONTACTING THE SHROUD HOUSING AND BECOMING DAMAGED.

(e) Turn the centerscrew (15) clockwise until the disk is released. This will be felt when the puller becomes free.

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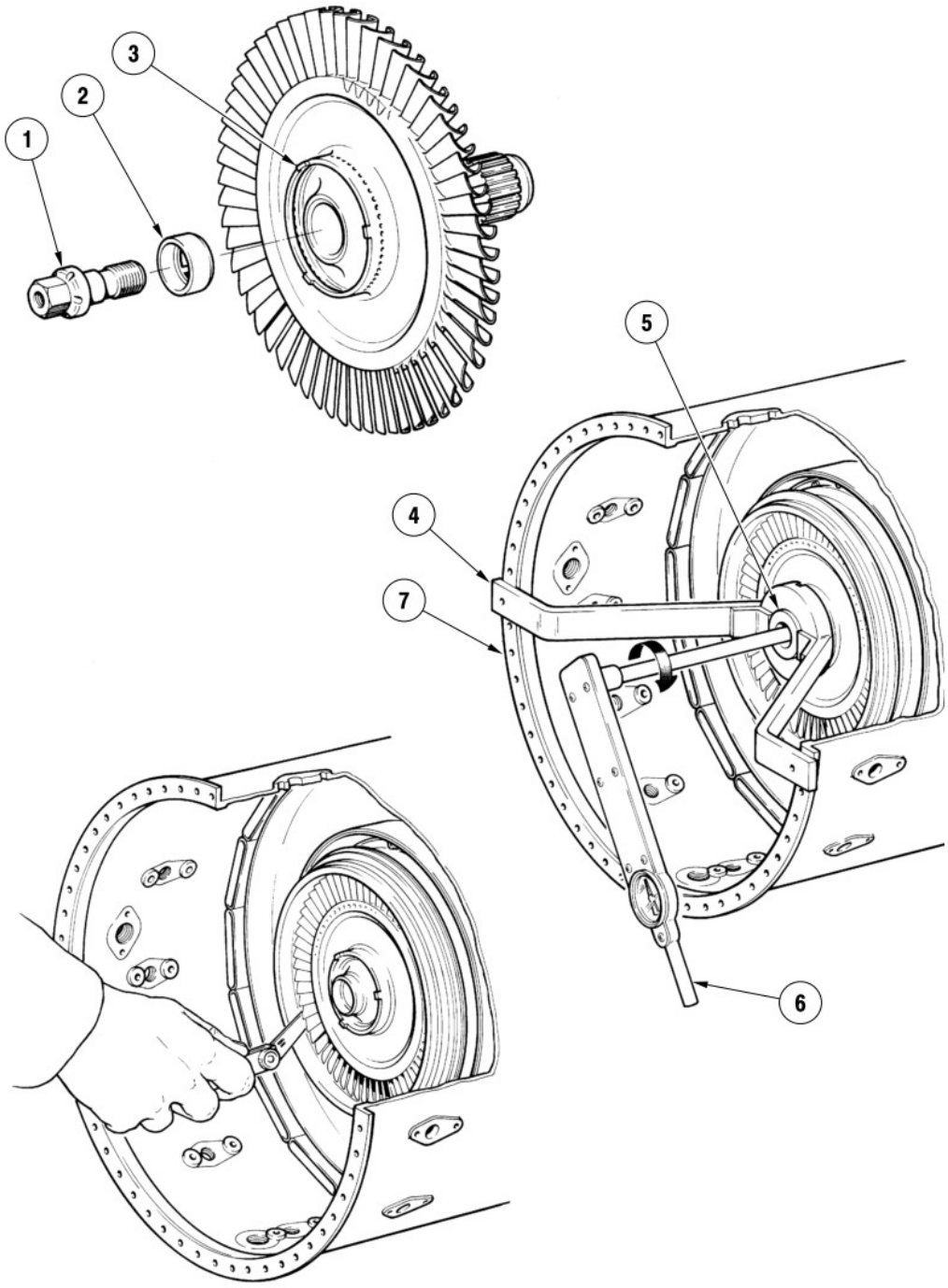
(13) Withdraw puller (10) and compressor turbine (7) from stubshaft and place on bench. Remove puller.

(14) Protect turbine in a suitable container.

B. Installation of Compressor Turbine Disk Assembly (Ref. Figs 203 and 204)

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Figure 203 Installation of Compressor Turbine and Turbine Blade Tip Clearance Check -Typical



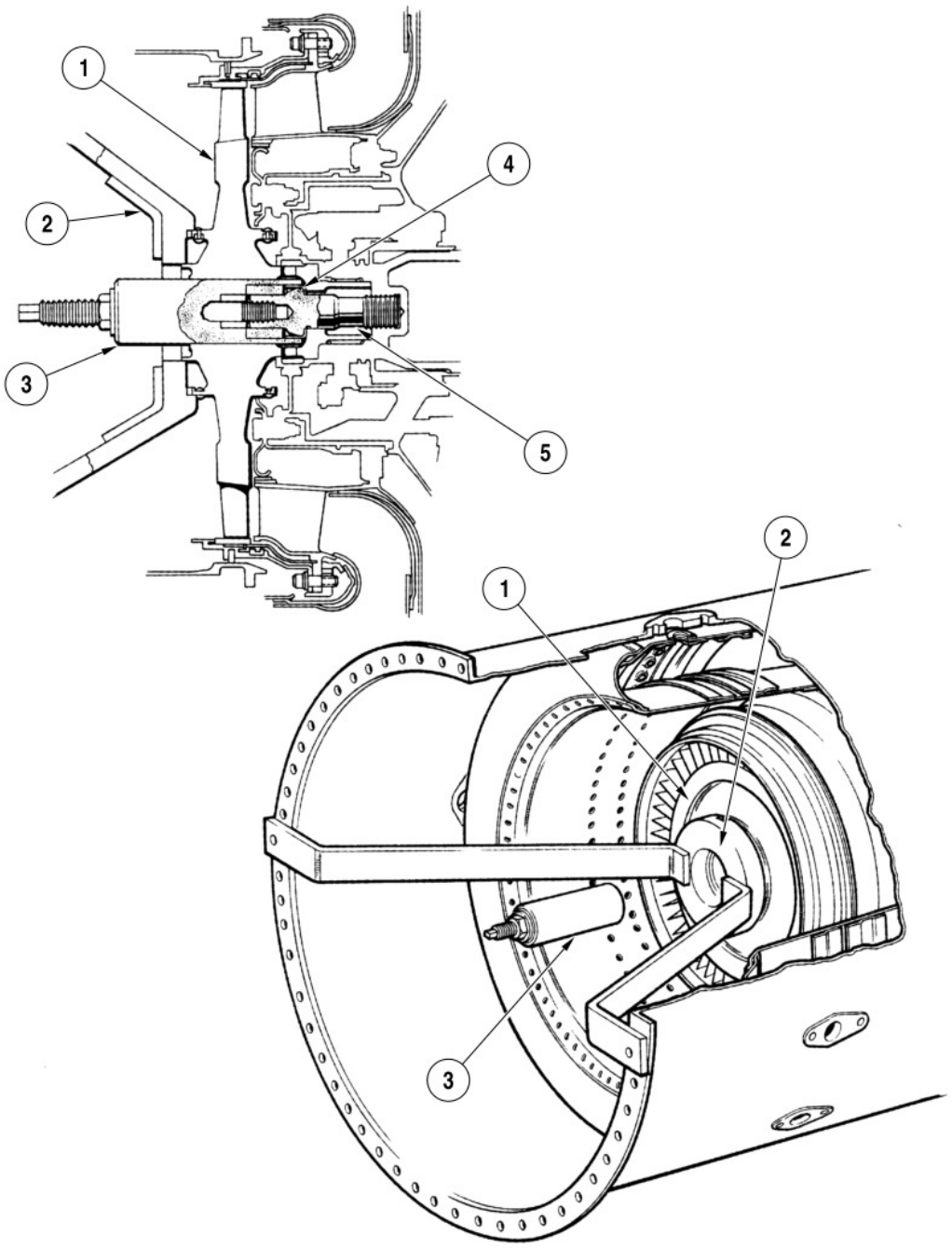
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1. Compressor Turbine Stubshaft Bolt
2. Keywasher
3. Compressor Turbine Disk Assembly
4. Wrench (PWC30331)
5. Protective Sleeve (PWC30336)
6. Torque Wrench
7. Gas Generator Case

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Figure 204 Locking Keywasher on Stubshaft Bolt



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1. Compressor Turbine Disk Assembly
2. Wrench (PWC30331)
3. Squeezer (PWC30458)
4. Keywasher
5. Stubshaft Bolt

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CAUTION: HANDLE THE TURBINE DISK AND BLADE ASSEMBLY WITH CARE.

(1) Apply a thin layer of antiseize compound (PWC06-004) to threads and shoulder of the stubshaft bolt (1, Fig. 203) and inside surface of the new keywasher (2). Make sure that the lubricating axial hole and slots in threads of the stubshaft bolt (Post-SB1439) are free of compound.

NOTE: Make sure that the antiseize compound (PWC06-004) is applied equally all over without any break.

(2) Align the master splines and install the compressor turbine (3) on the compressor front stubshaft.

(3) Place keywasher (2) on bolt (1). Locate bolt in turbine bore and engage threads in stubshaft.

(4) Place wrench (4) on gas generator case (7) with extension arm locating pins inserted into bolt holes in Flange C. Align disk lugs with wrench slots.

(5) Insert protector (5) into bore of wrench (4) and compressor turbine disk.

(6) Using a conventional socket wrench and torque wrench (6), tighten the stubshaft bolt (1):

(a) Initially tighten 600 to 650 lb.in.

(b) Slacken off to zero.

(c) For Pre-SB1439 Engines: Torque 420 to 460 lb.in.

(d) For Post-SB1439 Engines: Torque 500 to 550 lb.in.

NOTE: Refer to Chapter 72-50-01, Table 203 for REF. NO. 541.

(7) Remove torque wrench, socket, protector and wrench from gas generator case.

(8) Check CT blade radial tip clearance (Ref. Para. 6.A.).

(9) When the clearances are satisfactory, mount wrench (2, Fig. 204) on gas generator case (Ref. Step (4) preceding).

(10) Install the compressor turbine disk squeezer (3) into disk bore and engage center bolt with thread in stubshaft bolt head (5). Crimp keywasher (4) by turning hexagon nut until squeezer reaches maximum travel.

(11) Remove squeezer (3) and wrench (2). Check keywasher for correct locking.

(12) Remove the power section from stand (Ref. 72-00-00, REMOVAL/INSTALLATION).

(13) Install power section onto the gas generator assembly (Ref. 72-00-00, REMOVAL/INSTALLATION).

6. Inspection/Check

A. Compressor Turbine Blade Tip Clearance

NOTE: The turbine blade tip clearance should be measured after the installation of the compressor turbine disk and blade assembly and whenever the engine performance is unsatisfactory (Ref. Removal/Installation and Chap. 72-00-00, FAULT ISOLATION).

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(1) Measure and record the turbine tip clearance at several locations (Ref. Removal/Installation).

(2) The tip clearance at any one location should not normally exceed 0.022 inch or be less than 0.008 inch. However, if the clearance does exceed 0.022 inch at one location only, the shroud segments may remain in service provided the engine performance has been satisfactory.

(3) If the tip clearances are less than 0.008 inch due to a high spot in a shroud segment, the segment may be ground locally, as required, over a length not exceeding one inch on any one segment.

NOTE: For the grinding procedure, refer to 72-50-01, Approved Repairs.

(4) Examine segments for curling. Localized reduction of tip clearance at segment outer edges permitted, given that it must obey the requirement that follows (Ref. Fig. 205):

NOTE: If you find any segment that does not obey the necessary requirement, then grind the segment ends. Make sure that the grind length is not more than 1.0 in. (25.4 mm) from each segment end.

(a) There is no indication of blade rubbing.

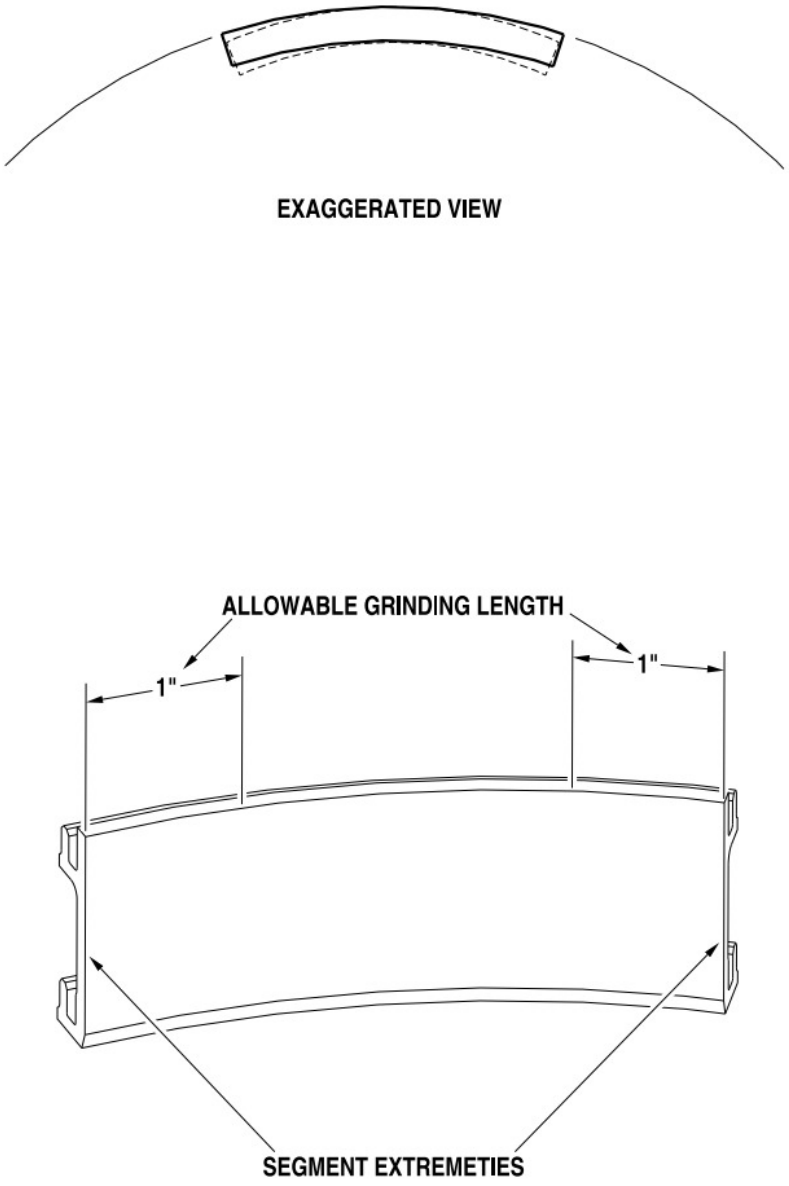
(b) The radial tip clearance at segment outer edge is 0.012 in. (0.304 mm) minimum.

(c) The radial tip clearance at segment center is in limits (REF. NO. 318, Fits and Clearances).

(5) Inspect the turbine shroud segments for evidence of metal buildup and cracks .

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Figure 205 Shroud Segment Curling



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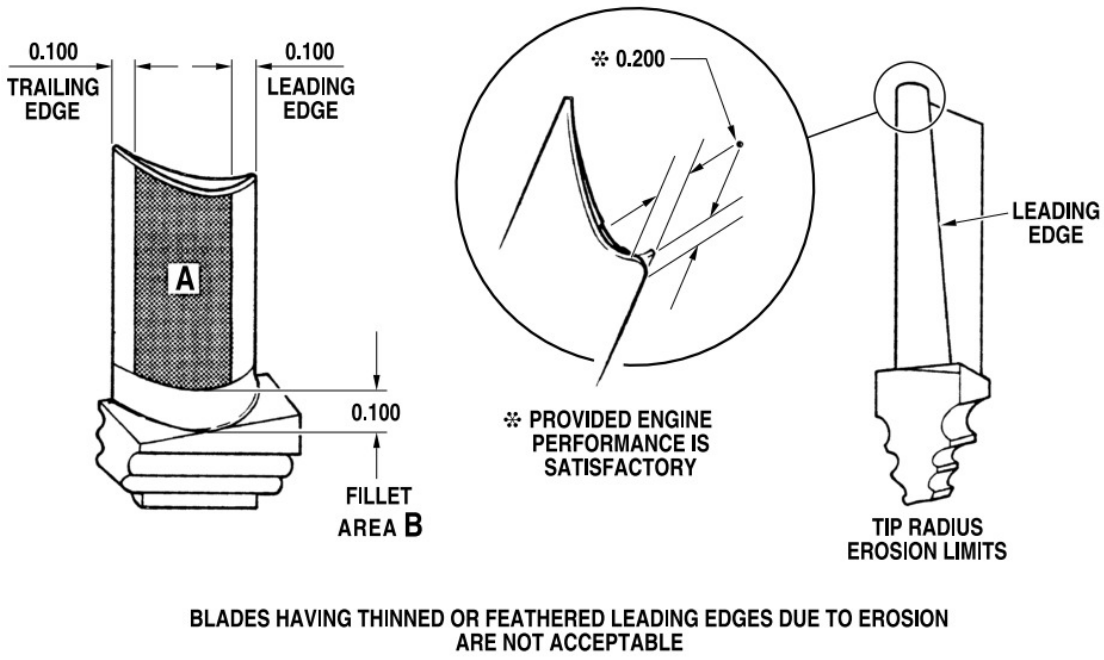
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B. Compressor Turbine Blades

(1) Inspect the compressor turbine blades (Ref. Table 201 and Fig. 206).

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Figure 206 Compressor Turbine Blade Inspection



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(2) The entire set of the compressor turbine blades must be scrapped when a compressor turbine vane ring inspection reveals that vane trailing edge burning has resulted in the loss of an airfoil area of 0.125 sq.in., or more, on any one vane and burnt area can be seen from the leading edge of the vane ring when viewed parallel to the longitudinal axis of the engine (Ref. 72-50-01, Table 201).

(3) Inspect blade airfoil surfaces, particularly concave surface for corrosion and loss of coating with 10-power magnification. Assess stage of deterioration and accept or reject bladed disk as follows (Ref. Fig. 207):

(a) Stage 1 - Initial Coating Deterioration (Sulfidation)

Evidenced by light color change of part of coating area, can be rust colored or dark gray. Coating has deteriorated but is probably still intact. Blades are acceptable for further service.

(b) Stage 2 - Initial Corrosion

Evidenced by apparent rise of sulfidated coating over surrounding surface, with small, scattered blisters appearing in coating. Corrosion of base material has started. Accept or reject disk, based on previous experience of the operator. If blades are to remain in service, operator must increase desalination washes and schedule regular borescope inspection of blades every 200 hours.

(c) Stage 3 - Advanced Corrosion

Evidenced by clusters of ruptured blisters that show bare material. Craters so formed deepen progressively, and crater surfaces darken with glazed appearance. Send the disk assembly to an approved P&WC overhaul facility for blade replacement in accordance with Overhaul Manual.

CAUTION: SEVERE CORROSION CAN RESULT IN IMBALANCE OF THE COMPRESSOR TURBINE ASSEMBLY AND SUBSEQUENT DAMAGE.

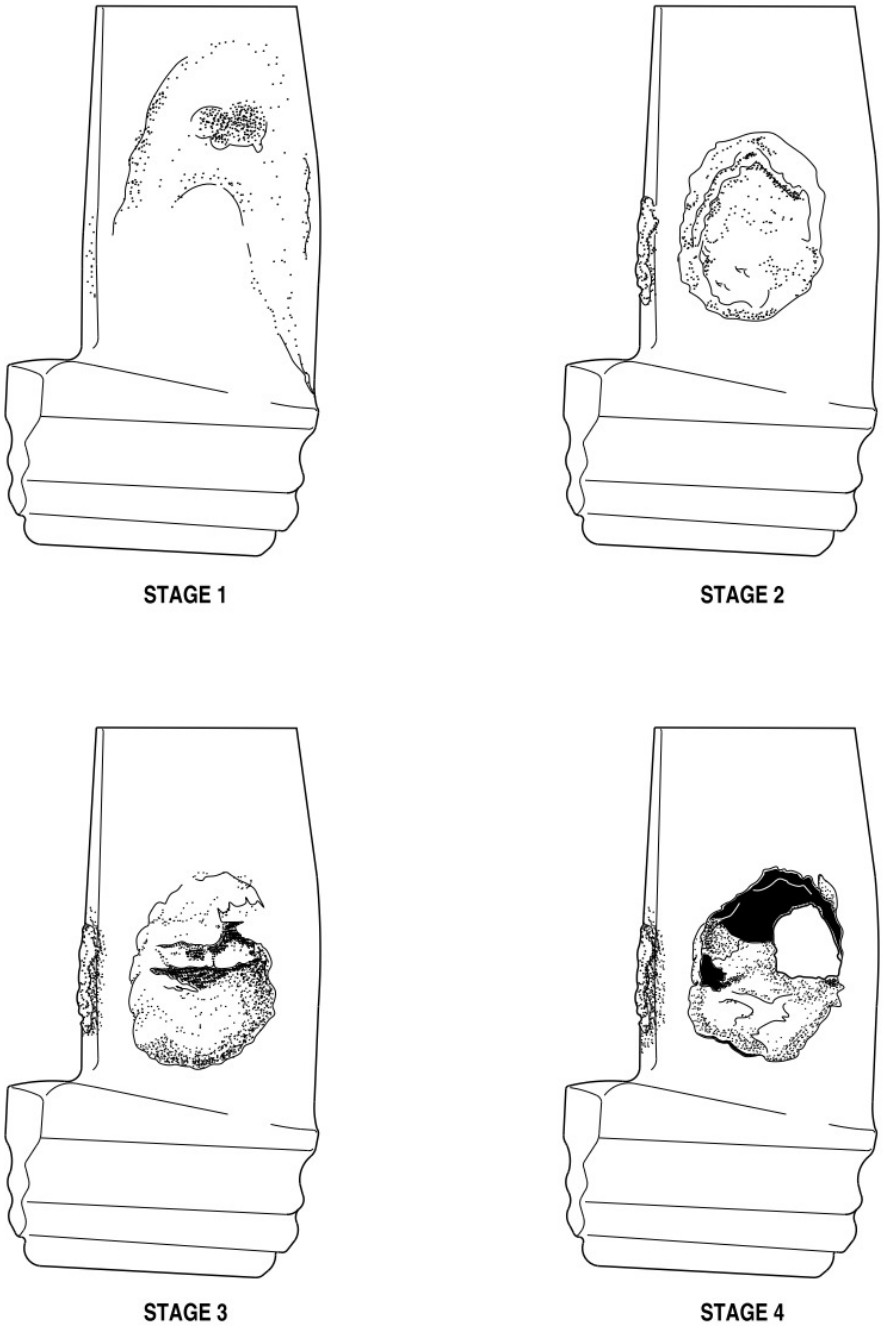
(d) Stage 4 - Severe Corrosion

Deep penetration with large ruptured blisters that show large areas of bare material. Send the power section to an approved P&WC overhaul facility for the disk repair, and inspection of No. 2 bearing area in accordance with Overhaul Manual.

NOTE: Corrosion can be identified as metal loss or pitting, but more usually it appears as local swelling or build-up due to greater volume occupied by the nickel oxides. These condition products vary in color from green to black and in the advanced state there will be associated flaking. Care must be taken to identify the difference between corrosion build-up and possible light brown or rust colored deposits which are essentially harmless combustion by-products. The latter are more widespread over hot section components and while possibly affect the performance, will not directly affect the blade integrity.

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Figure 207 Stages of corrosion



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Table 201 Compressor Turbine Blade Inspection

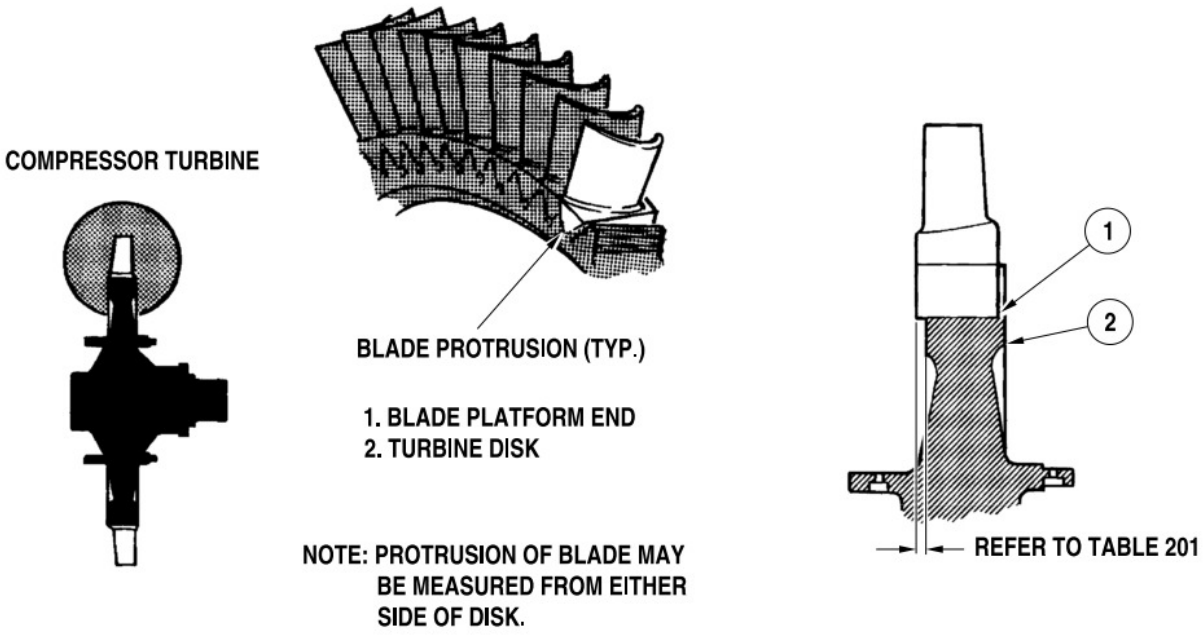
Inspection	**Maximum Serviceable**	**Corrective Action**
CORROSION (Ref. 72-00-00)	Inspect blade airfoil surfaces, paticularily concave surface for corrosion and loss of coating using a 10X magnification. A loss of up to 25% of coating is permissible. Assess stage of deterioration and accept or reject bladed disk.	Not repairable
BLADE SHIFT	Protrusion of blade root beyond disk rim must be equal within 0.016 inch on either side of disk (Ref. Fig. 208).	Ref. NOTE 1
CIRCUMFERENTIAL MOVEMENT	Max. 0.030 inch acceptable at blade tip	
SQUEALER TIP RUB	Minimum squealer tip height 0.008 inch.	Ref. NOTE 1
AREA A		
Nicks, dents and pits	Isolated pits up to 0.008 inch deep maximum, not more than 30% of the total area. Maximum six nicks or dents per square inch, up to 0.015 inch deep and 0.100 inch apart.	Ref. NOTE 1
Cracks	Not permitted	Ref. NOTE 2
AREA B		
Nicks, dents and pits	Isolated pits up to 0.008 inch deep maximum, not more than 30% of the total area. Maximum four nicks or dents per square inch, up to 0.010 inch deep and 0.100 inch apart.	Ref. NOTE 1
Cracks	Not permitted	Ref. NOTE 2
LEADING EDGE		
Nicks, dents and pits	Isolated pits up to 0.008 inch deep maximum, not more than 30% of the total area. Maximum four nicks or dents per square inch, up to 0.020 inch deep and 0.100 inch apart.	Ref. NOTE 1
Cracks	Not permitted	Ref. NOTE 2

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Inspection TRAILING EDGE	**Maximum Serviceable**	**Corrective Action**
Nicks, dents and pits	Isolated pits up to 0.005 inch deep maximum, not more than 30% of the total area. Maximum four nicks or dents per square inch, up to 0.010 inch deep and 0.100 inch apart.	Ref. NOTE 1
Cracks	Not permitted	Ref. NOTE 2
LEADING EDGE TIP		
Erosion	Erosion up to a maximum radius of 0.200 in.	Ref. NOTE 1
NOTE: 1. Replacement of individual blades is not authorized. Send bladed disk assembly to an approved overhaul facility.		
NOTE: 2. If any blade is found cracked, return turbine assembly to an approved overhaul facility for replacement of complete set of blades.		
NOTE: 3. Cracks may not extend beyond coating depth. If, after stripping coating, the cracks do not extend into the base metal, the blades may be considered serviceable. Other, normal inspection criteria must be applied.		

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Figure 208 Compressor Turbine Blade Shift



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C. Compressor Turbine Stubshaft Bolt (Post-SB1439)

(1) Examine the stubshaft bolt lubricating axial hole and the two slots in the threads to make sure these areas are clear of compound and debris.

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